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 Studierichting: Informatica
 Oefeningenreeks 1D nr. 23.3

Los de volgende vergelijking op in $\mathbb{C}$:

$$z^3 + (4 + 2i)z^2 + (5 + 6i)z + 2 + 4i = 0$$

$$A(-1) = -1 + 4 + 2i - 5 - 6i + 2 + 4i = 0$$

$$\begin{array}{c|ccc|c} -1 & 1 & 4+2i & 5+6i & 2+4i \\ & & -1 & -3-2i & -2-4i \\ \hline & 1 & 3+2i & 2+4i & 0 \end{array}$$

$$\Leftrightarrow (z+1)(z^2 + (3+2i)z + 2+4i) = 0$$

$$\Delta = (3+2i)^2 - 4(2+4i) = (9+12i-4) - (8+16i) = -3-4i$$

$$\Leftrightarrow \begin{cases} x^2 - y^2 = -3 \\ 2xy = -4 \end{cases}$$

$$\Leftrightarrow \begin{cases} x^2 - y^2 = -3 \\ x^2(-y)^2 = 4 \end{cases}$$

$$\lambda^2 + 3\lambda - 4 = 0$$

$$\Delta_\lambda = 3^2 - 4 * (-4) = 25$$

$$\Leftrightarrow \lambda = \frac{-3 \pm \sqrt{25}}{2} = \frac{-3 \pm 5}{2}$$

$$\Leftrightarrow \lambda_1 = 1 \text{ en } \lambda_2 = -4$$

$$\Leftrightarrow x = \pm 1 \text{ en } y = \pm 2$$

$$\Leftrightarrow k_1 = 1 - 2i \text{ en } k_2 = -1 + 2i$$

$$\Leftrightarrow z_3 = \frac{-3 - 2i + 1 - 2i}{2} = -1 - 2i \text{ en } z_4 = \frac{-3 - 2i - 1 + 2i}{2} = -2$$

$$\text{Nulpunten: } \{-1, -2, -1 - 2i\}$$

References